

Air University



## SSDD (Lab)

Year 2026

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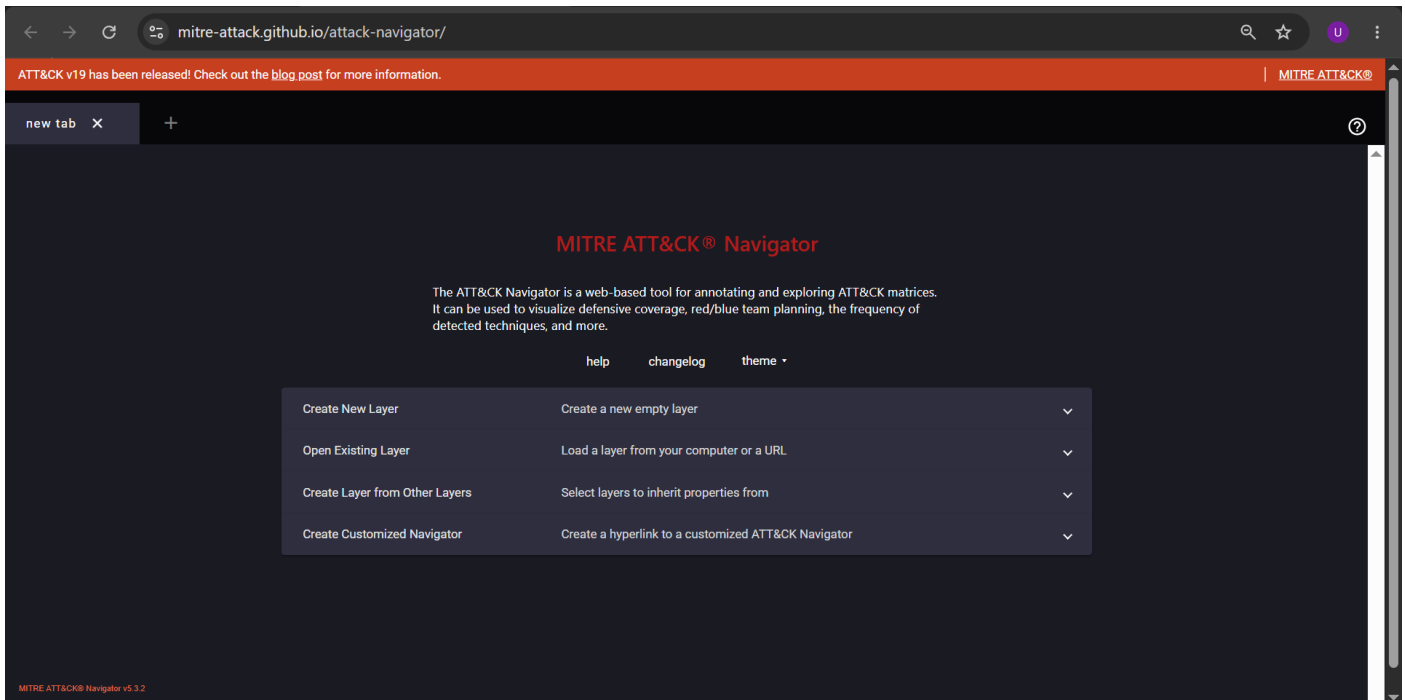
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Department of Cyber Security

USAMA ARSHAD | 247141  
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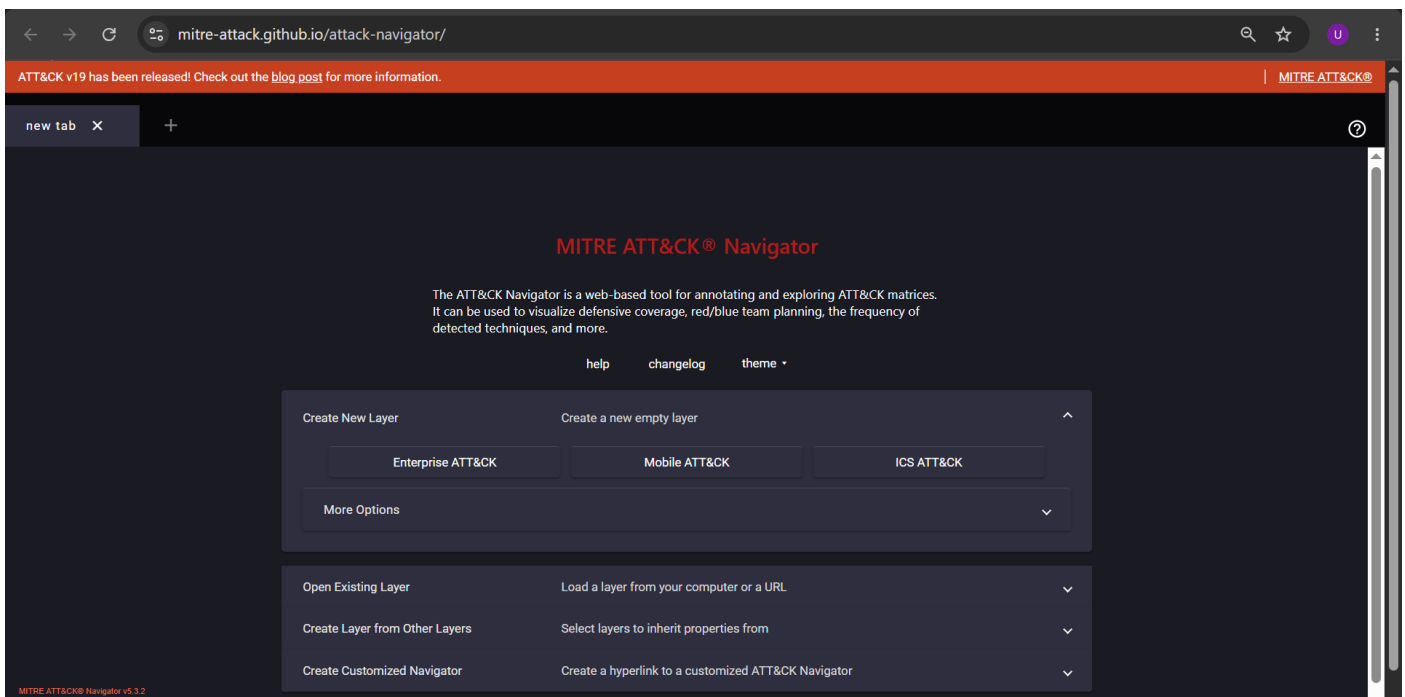
## Lab # 10

### PART 1 — Setting Up ATT&CK Navigator

#### ATT&CK Navigator — Landing Page



#### Creating New Layer — Enterprise Matrix



## PART 2 — TASK 1: Scenario 1 (SecureBank — Enterprise Matrix)

### Step A: Know Exactly What to Highlight

| # | Security Flaw                        | Tactic            | Technique ID | Technique Name                    |
|---|--------------------------------------|-------------------|--------------|-----------------------------------|
| 1 | SMS-based MFA → SIM Swapping         | Credential Access | T1078        | Valid Accounts                    |
| 2 | Unencrypted backups                  | Impact            | T1485        | Data Destruction                  |
| 3 | Basic antivirus (fileless malware)   | Defense Evasion   | T1055        | Process Injection                 |
| 4 | Quarterly patching (zero-day window) | Persistence       | T1078        | Valid Accounts                    |
| 5 | SQL injection / XSS on web app       | Initial Access    | T1190        | Exploit Public-Facing Application |

**Note:** T1078 appears twice (for flaws 1 and 4) under different tactics. This is valid in ATT&CK — one technique can serve multiple tactics.

### Step B: Selecting Techniques in the Navigator

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layer x +

Selection Controls Layer Controls Technique Controls

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| Reconnaissance                           | Resource Development            | Initial Access                            | Execution                                | Persistence                                | Privilege Escalation                       | Stealth                                       | Defense Impairment                                | Credential Access                         | Discovery                              | Lateral Movement                            | Collection                               | Command and Control                         |
|------------------------------------------|---------------------------------|-------------------------------------------|------------------------------------------|--------------------------------------------|--------------------------------------------|-----------------------------------------------|---------------------------------------------------|-------------------------------------------|----------------------------------------|---------------------------------------------|------------------------------------------|---------------------------------------------|
| 12 techniques                            | 9 techniques                    | 11 techniques                             | 20 techniques                            | 22 techniques                              | 13 techniques                              | 30 techniques                                 | 18 techniques                                     | 17 techniques                             | 34 techniques                          | 9 techniques                                | 17 techniques                            | 18 techniques                               |
| Active Scanning (0/3)                    | Acquire Access (0/3)            | Content Injection (0/3)                   | BITS Jobs (0/3)                          | Account Manipulation (0/7)                 | Abuse Elevation Control Mechanism (0/6)    | Access Token Manipulation (0/5)               | Disable or Modify System Firewall (0/3)           | Adversary-in-the-Middle (0/4)             | Account Discovery (0/4)                | Exploitation of Remote Services (0/4)       | Adversary-in-the-Middle (0/4)            | Application Layer Protocol (0/3)            |
| Gather Victim Host Information (0/4)     | Acquire Infrastructure (0/8)    | Drive-by Compromise (0/3)                 | Cloud Administration Command (0/13)      | BITS Jobs (0/7)                            | Build Image on Host (0/6)                  | Debugger Evasion (0/6)                        | Disable or Modify Tools (0/4)                     | Brute Force (0/4)                         | Application Window Discovery (0/4)     | Internal Spearphishing (0/4)                | Archive Collected Data (0/3)             | Communication Through Removable Media (0/3) |
| Gather Victim Identity Information (0/3) | Compromise Accounts (0/3)       | Exploit Public-Facing Application (0/13)  | Command and Scripting Interpreter (0/13) | Boot or Logon Autostart Execution (0/14)   | Access Token Manipulation (0/5)            | Delay Execution (0/2)                         | Domain or Tenant Policy Modification (0/2)        | Credentials from Password Stores (0/6)    | Browser Information Discovery (0/6)    | Lateral Tool Transfer (0/2)                 | Audio Capture (0/2)                      | Content Injection (0/2)                     |
| Gather Victim Network Information (0/6)  | Compromise Infrastructure (0/8) | External Remote Services (0/4)            | Container Administration Command (0/13)  | Boot or Logon Initialization Scripts (0/5) | Account Manipulation (0/7)                 | Deobfuscate/Decode Files or Information (0/2) | Downgrade Attack (0/2)                            | Exploitation for Credential Access (0/2)  | Cloud Infrastructure Discovery (0/2)   | Remote Service Hijacking (0/2)              | Automated Collection (0/2)               | Data Encoding (0/2)                         |
| Gather Victim Org Information (0/4)      | Develop Capabilities (0/4)      | Hardware Additions (0/4)                  | Deploy Container (0/14)                  | Cloud Application Integration (0/14)       | Boot or Logon Autostart Execution (0/7)    | Execution Guardrails (0/2)                    | Exploitation for Defense Impairment (0/2)         | Forced Authentication (0/2)               | Cloud Service Dashboard (0/2)          | Remote Services (0/8)                       | Browser Session Hijacking (0/3)          | Data Obfuscation (0/3)                      |
| Phishing for Information (0/4)           | Establish Accounts (0/3)        | Phishing (0/4)                            | ESXi Administration Command (0/12)       | Compromise Host Software Binary (0/3)      | Boot or Logon Initialization Scripts (0/5) | Exploitation for Stealth (0/14)               | File and Directory Permissions Modification (0/2) | Forge Web Credentials (0/2)               | Cloud Service Discovery (0/2)          | Replication Through Removable Media (0/2)   | Clipboard Data (0/2)                     | Dynamic Resolution (0/3)                    |
| Query Public AI Services (0/2)           | Generate Content (0/2)          | Replication Through Removable Media (0/3) | Exploitation for Client Execution (0/12) | Create Account (0/3)                       | Create or Modify System Process (0/5)      | Hide Artifacts (0/14)                         | Input Capture (0/4)                               | Modify Authentication Process (0/9)       | Container and Resource Discovery (0/2) | Software Deployment Tools (0/2)             | Data from Cloud Storage (0/2)            | Encrypted Channel (0/2)                     |
| Search Closed Sources (0/2)              | Obtain Capabilities (0/7)       | Supply Chain Compromise (0/3)             | Hijack Execution Flow (0/12)             | Create or Modify System Process (0/5)      | Domain or Tenant Policy Modification (0/2) | Hijack Execution Flow (0/12)                  | Indicator Removal (0/8)                           | Multi-Factor Authentication Process (0/9) | Debugger Evasion (0/2)                 | Taint Shared Content (0/2)                  | Data from Configuration Repository (0/2) | Fallback Channels (0/2)                     |
| Search Open Technical Databases (0/5)    | Stage Capabilities (0/3)        | Trusted Relationship (0/3)                | Inter-Process Communication (0/3)        | Event Triggered Execution (0/2)            | Obfuscated Files or Information (0/2)      | Indirect Command Execution (0/12)             | Masquerading (0/12)                               | Modify Cloud Compute Infrastructure (0/2) | Device Driver Discovery (0/2)          | Use Alternate Authentication Material (0/4) | Data from Information Repositories (0/2) | Ingress Tool Transfer (0/2)                 |
| Search Open Websites/Domains (0/3)       |                                 |                                           |                                          |                                            |                                            |                                               |                                                   |                                           | File and Directory Discovery (0/2)     |                                             |                                          |                                             |

legend

The screenshot displays the MITRE ATT&CK v9 interface. The top navigation bar includes 'Selection Controls', 'Layer Controls', and 'Technique Controls'. The main content area is divided into columns representing different stages of an attack: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, and Defense Impairment. The 'Valid Accounts' technique is highlighted in the 'Persistence' column. The right sidebar shows search results for 'T1078', including search settings and a list of techniques with 'Valid Accounts' selected. The bottom section shows threat groups and mitigations.

The screenshot displays the MITRE ATT&CK v9 interface. The top navigation bar includes 'Selection Controls', 'Layer Controls', and 'Technique Controls'. The main content area is divided into columns representing different stages of an attack: Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact. The 'Data Destruction' technique is highlighted in the 'Impact' column. The right sidebar shows search results for 'T1485', including search settings and a list of techniques with 'Data Destruction' selected. The bottom section shows threat groups and mitigations.

Selection Controls | Layer Controls | Technique Controls

Search: T1055

Search Settings: ☐ Name ☒ ATT&CK ID ☐ Description

Techniques (13)

select all deselect all

Process Injection [view](#) select deselect

Process Injection : Asynchronous Procedure Call [view](#) select deselect

Process Injection : Dynamic-link Library Injection [view](#) select deselect

Process Injection : Extra Window Memory Injection [view](#) select deselect

Threat Groups (0)

Software (0)

Mitigations (0)

legend

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layer X +

Selection Controls | Layer Controls | Technique Controls

Search: T1190

Search Settings: ☐ Name ☒ ATT&CK ID ☐ Description

Techniques (1)

select all deselect all

Exploit Public-Facing Application [view](#) select deselect

Threat Groups (0)

Software (0)

Mitigations (0)

Campaigns (0)

legend

## Step C: Highlighting Techniques

The screenshot displays the MITRE ATT&CK v19 interface with the 'Stealth' layer selected. The interface is organized into columns representing different stages of an attack: Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, and Credential Access. The 'Stealth' column contains 30 techniques, with several highlighted in orange: Asynchronous Procedure Call, Dynamic-link Library Injection, Extra Window Memory Injection, ListPlanting, Portable Executable Injection, Proc Memory, Process Doppelgänger, Process Hollowing, Ptrace System Calls, Thread Execution Hijacking, Thread Local Storage, and VDSO Hijacking. The right-hand panel shows the 'Techniques (13)' list with a color selection tool and a 'Threat Groups (0)' section.

The screenshot displays the MITRE ATT&CK v19 interface with the 'Initial Access' layer selected. The interface is organized into columns representing different stages of an attack: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, and Credential Access. The 'Initial Access' column contains 11 techniques, with several highlighted in red: Acquire Access, Acquire Infrastructure, Compromise Accounts, Compromise Infrastructure, Develop Capabilities, Establish Accounts, Generate Content, Obtain Capabilities, Stage Capabilities, and Trusted Relationship. The right-hand panel shows the 'Techniques (1)' list with a search bar and a 'Threat Groups (0)' section.

Selection Controls | Layer Controls | Technique Controls

Reconnaissance 12 techniques | Resource Development 9 techniques | Initial Access 11 techniques | Execution 20 techniques | Persistence 22 techniques | Privilege Escalation 13 techniques | Stealth 30 techniques

Search: T1078

Search Settings: Name, ATT&CK ID

Techniques (5): Valid Accounts, Valid Accounts: Cloud Accounts, Valid Accounts: Default Accounts, Valid Accounts: Domain Accounts, Valid Accounts: Mitigations

Threat Groups (0)

Software (0)

Mitigations (0)

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layer x +

Selection Controls | Layer Controls | Technique Controls

Search: T1485

Search Settings: Name, ATT&CK ID

Techniques (2): Data Destruction, Data Destruction: Lifecycle-Triggered Deletion

Threat Groups (0)

Software (0)

Mitigations (0)

## Step D: Adding Score

The screenshot displays the MITRE ATT&CK v9.1 interface, specifically the 'Stealth' layer. The interface is divided into several sections: a top navigation bar with 'Selection Controls', 'Layer Controls', and 'Technique Controls'; a main content area with a grid of techniques; and a right-hand sidebar with search and filter options.

**Techniques (5):**

- Valid Accounts: [view](#) [select](#) [deselect](#)
- Valid Accounts : Cloud Accounts: [view](#) [select](#) [deselect](#)
- Valid Accounts : Default Accounts: [view](#) [select](#) [deselect](#)
- Valid Accounts : Domain Accounts: [view](#) [select](#) [deselect](#)

**Threat Groups (0)**

**Software (0)**

**Mitigations (0)**

**Legend**

The screenshot displays the MITRE ATT&CK v9.1 interface, specifically the 'Impact' layer. The interface is divided into several sections: a top navigation bar with 'Selection Controls', 'Layer Controls', and 'Technique Controls'; a main content area with a grid of techniques; and a right-hand sidebar with search and filter options.

**Techniques (2):**

- Data Destruction: [view](#) [select](#) [deselect](#)
- Data Destruction : Lifecycle-Triggered Deletion: [view](#) [select](#) [deselect](#)

**Threat Groups (0)**

**Software (0)**

**Mitigations (0)**

**Legend**

Selection Controls | Layer Controls | Technique Controls

score 9

Search Settings

☐ Name ☒ ATT&CK ID ☐ Description

Techniques (13)

select all deselect all

Process Injection [view](#) [select](#) [deselect](#)

Process Injection : Asynchronous Procedure Call [view](#) [select](#) [deselect](#)

Process Injection : Dynamic-link Library Injection [view](#) [select](#) [deselect](#)

Process Injection : Extra Window Memory Injection [view](#) [select](#) [deselect](#)

Threat Groups (0)

Software (0)

Mitigations (0)

legend

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layer x +

Selection Controls | Layer Controls | Technique Controls

score 10

Search Settings

☐ Name ☒ ATT&CK ID ☐ Description

Techniques (1)

select all deselect all

Exploit Public-Facing Application [view](#) [select](#) [deselect](#)

Threat Groups (0)

Software (0)

Mitigations (0)

Campaigns (0)

legend

## Step E: Adding Annotations

The screenshot displays the MITRE ATT&CK v9 interface. The main grid shows various techniques categorized into layers: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, and Credential Access. The 'Valid Accounts' technique (T1078) is highlighted in the 'Initial Access' layer. The right sidebar shows search results for 'T1078' and a list of techniques.

Selection Controls | Layer Controls | Technique Controls

Search: T1078

Search Settings: ☐ Name ☒ ATT&CK ID ☐ Description

Techniques (5)

- Valid Accounts: Cloud Accounts [view](#) [select](#) [deselect](#)
- Valid Accounts: Default Accounts [view](#) [select](#) [deselect](#)
- Valid Accounts: Domain Accounts [view](#) [select](#) [deselect](#)
- Valid Accounts: Local Accounts [view](#) [select](#) [deselect](#)

Threat Groups (0)

Software (0)

Mitigations (0)

legend

## Step F: Export the Matrix

The screenshot displays the MITRE ATT&CK v9 interface. The main grid shows various techniques categorized into layers: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, and Credential Access. The 'Valid Accounts' technique (T1078) is highlighted in the 'Initial Access' layer. The right sidebar shows search results for 'T1078' and a list of techniques.

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layer X +

Selection Controls | Layer Controls | Technique Controls

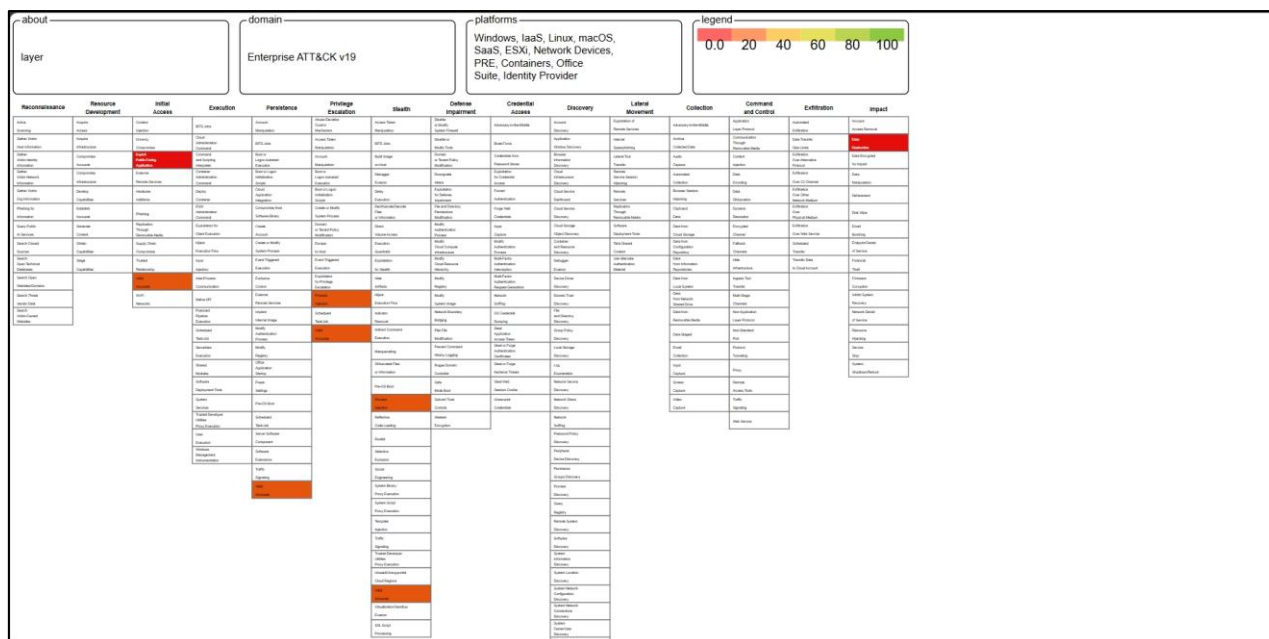
Export: [Mc](#) [<>](#) [Grid](#) [Image](#) [PDF](#) [SVG](#) [JSON](#) [YAML](#) [CSV](#) [XML](#) [HTML](#) [Markdown](#) [Text](#) [Image](#) [PDF](#) [SVG](#) [JSON](#) [YAML](#) [CSV](#) [XML](#) [HTML](#) [Markdown](#) [Text](#)

Render layer to SVG

Options: ☐ Show technique details on visible techniques

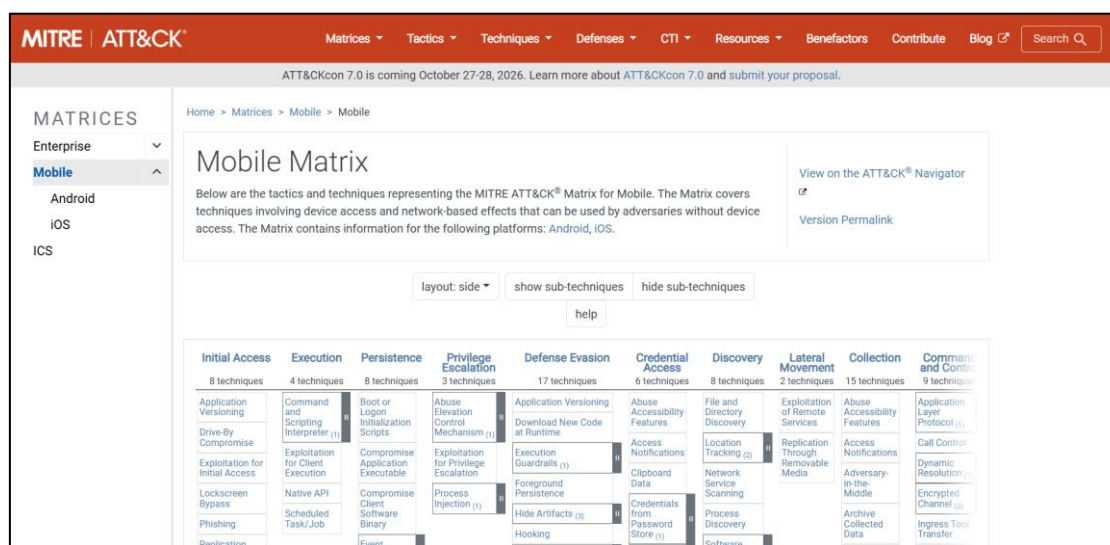
legend

## Step G: Results

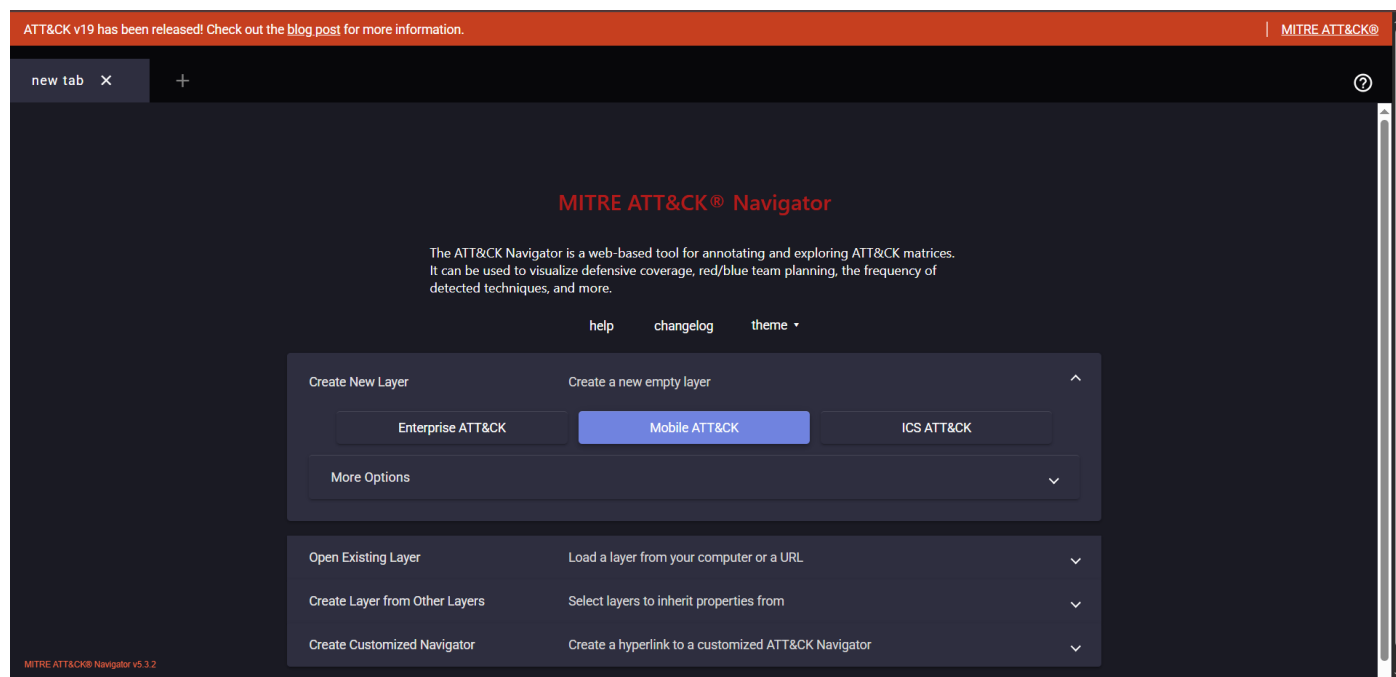


## PART 3 — TASK 2: Scenario 2 (MobilePay — Mobile Matrix)

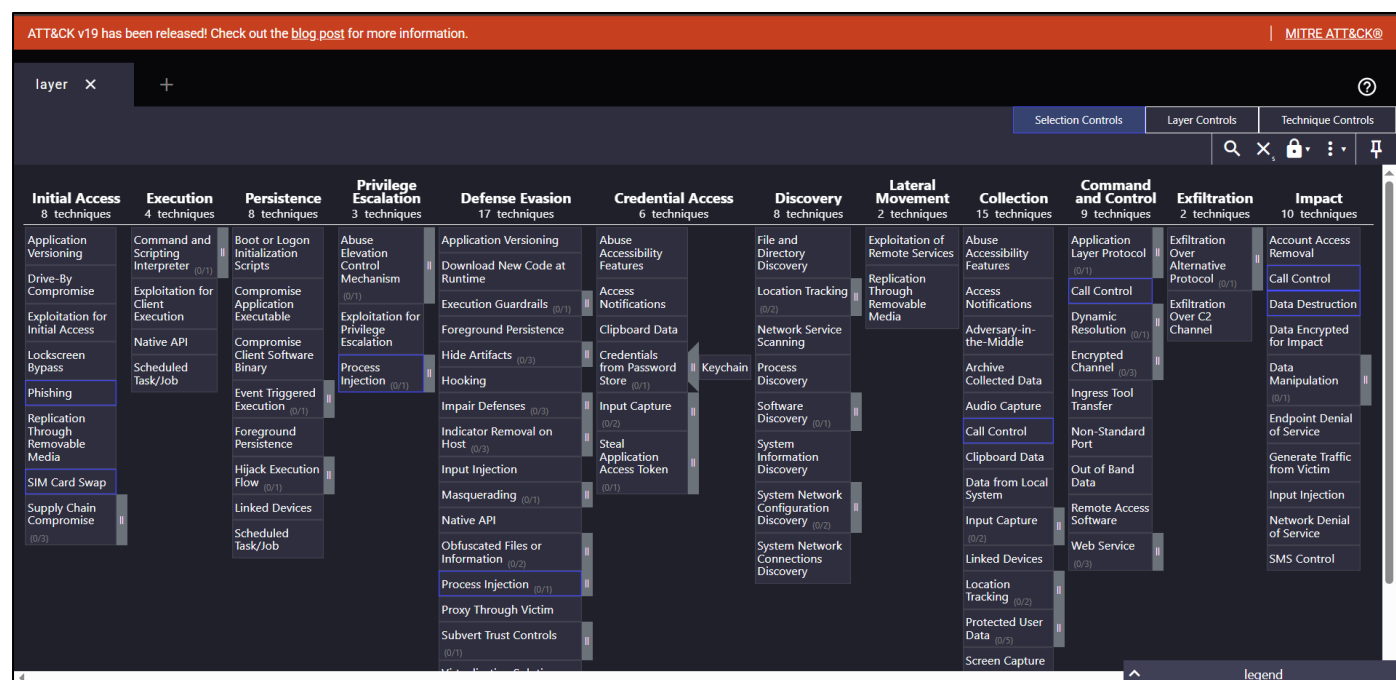
## Step A: Mobile Matrix



## Step B: Create Mobile Layer



## Step C: Add Techniques



## Step D: Add Colors

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layer × +

Selection Controls Layer Controls Technique Controls

| Initial Access                      | Execution                               | Persistence                          | Privilege Escalation                    | Defense Evasion                       | Credential Access                     | Discovery                                    | Lateral Movement                    | Collection                   | Command and Control              | Exfiltration                                 | Impact                       |
|-------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------------|-------------------------------------|------------------------------|----------------------------------|----------------------------------------------|------------------------------|
| 8 techniques                        | 4 techniques                            | 8 techniques                         | 3 techniques                            | 17 techniques                         | 6 techniques                          | 8 techniques                                 | 2 techniques                        | 15 techniques                | 9 techniques                     | 2 techniques                                 | 10 techniques                |
| Application Versioning              | Command and Scripting Interpreter (0/1) | Boot or Logon Initialization Scripts | Abuse Elevation Control Mechanism (0/1) | Application Versioning                | Abuse Accessibility Features          | File and Directory Discovery                 | Exploitation of Remote Services     | Abuse Accessibility Features | Application Layer Protocol (0/1) | Exfiltration Over Alternative Protocol (0/1) | Account Access Removal       |
| Drive-By Compromise                 | Exploitation for Client Execution       | Compromise Application Executable    | Exploitation for Privilege Escalation   | Download New Code at Runtime          | Access Notifications                  | Location Tracking (0/2)                      | Replication Through Removable Media | Access Notifications         | Call Control                     | Exfiltration Over C2 Channel                 | Call Control                 |
| Exploitation for Initial Access     | Native API                              | Compromise Client Software Binary    | Process Injection                       | Execution Guardrails (0/1)            | Clipboard Data                        | Network Service Scanning                     |                                     | Adversary-in-the-Middle      | Dynamic Resolution (0/1)         |                                              | Data Encrypted for Impact    |
| Lockscreen Bypass                   | Scheduled Task/Job                      | Event Triggered Execution (0/1)      |                                         | Foreground Persistence                | Credentials from Password Store (0/1) | Process Discovery                            |                                     | Archive Collected Data       | Encrypted Channel (0/2)          |                                              | Data Manipulation (0/1)      |
| Phishing                            |                                         | Foreground Persistence               |                                         | Hide Artifacts (0/2)                  | Input Capture (0/2)                   | Software Discovery (0/1)                     |                                     | Audio Capture                | Ingress Tool Transfer            |                                              | Endpoint Denial of Service   |
| Replication Through Removable Media |                                         | Hijack Execution Flow (0/1)          |                                         | Indicator Removal on Host (0/2)       | Steal Application Access Token (0/1)  | System Information Discovery                 |                                     | Call Control                 | Non-Standard Port                |                                              | Generate Traffic from Victim |
| SIM Card Swap                       |                                         | Linked Devices                       |                                         | Input Injection                       | Masquerading (0/1)                    | System Network Configuration Discovery (0/2) |                                     | Clipboard Data               | Out of Band Data                 |                                              | Input Injection              |
| Supply Chain Compromise (0/3)       |                                         | Scheduled Task/Job                   |                                         | Masquerading (0/1)                    |                                       | System Network Connections Discovery         |                                     | Data from Local System       | Remote Access Software           |                                              | Network Denial of Service    |
|                                     |                                         |                                      |                                         | Native API                            |                                       |                                              |                                     | Input Capture (0/2)          | Web Service (0/3)                |                                              | SMS Control                  |
|                                     |                                         |                                      |                                         | Obfuscated Files or Information (0/2) |                                       |                                              |                                     | Linked Devices               |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Process Injection (0/2)               |                                       |                                              |                                     | Location Tracking (0/2)      |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Proxy Through Victim                  |                                       |                                              |                                     | Protected User Data (0/1)    |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Subvert Trust Controls (0/1)          |                                       |                                              |                                     | Screen Capture               |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Virtualization Solution               |                                       |                                              |                                     |                              |                                  |                                              |                              |

legend

## Step E: Assign Scores

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layer × +

Selection Controls Layer Controls Technique Controls

| Initial Access                      | Execution                               | Persistence                          | Privilege Escalation                    | Defense Evasion                       | Credential Access                     | Discovery                                    | Lateral Movement                    | Collection                   | Command and Control              | Exfiltration                                 | Impact                       |
|-------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------------|-------------------------------------|------------------------------|----------------------------------|----------------------------------------------|------------------------------|
| 8 techniques                        | 4 techniques                            | 8 techniques                         | 3 techniques                            | 17 techniques                         | 6 techniques                          | 8 techniques                                 | 2 techniques                        | 15 techniques                | 9 techniques                     | 2 techniques                                 | 10 techniques                |
| Application Versioning              | Command and Scripting Interpreter (0/1) | Boot or Logon Initialization Scripts | Abuse Elevation Control Mechanism (0/1) | Application Versioning                | Abuse Accessibility Features          | File and Directory Discovery                 | Exploitation of Remote Services     | Abuse Accessibility Features | Application Layer Protocol (0/1) | Exfiltration Over Alternative Protocol (0/1) | Account Access Removal       |
| Drive-By Compromise                 | Exploitation for Client Execution       | Compromise Application Executable    | Exploitation for Privilege Escalation   | Download New Code at Runtime          | Access Notifications                  | Location Tracking (0/2)                      | Replication Through Removable Media | Access Notifications         | Call Control                     | Exfiltration Over C2 Channel                 | Call Control                 |
| Exploitation for Initial Access     | Native API                              | Compromise Client Software Binary    | Process Injection                       | Execution Guardrails (0/1)            | Clipboard Data                        | Network Service Scanning                     |                                     | Adversary-in-the-Middle      | Dynamic Resolution (0/1)         |                                              | Data Encrypted for Impact    |
| Lockscreen Bypass                   | Scheduled Task/Job                      | Event Triggered Execution (0/1)      |                                         | Foreground Persistence                | Credentials from Password Store (0/1) | Process Discovery                            |                                     | Archive Collected Data       | Encrypted Channel (0/2)          |                                              | Data Manipulation (0/1)      |
| Phishing                            |                                         | Foreground Persistence               |                                         | Hide Artifacts (0/2)                  | Input Capture (0/2)                   | Software Discovery (0/1)                     |                                     | Audio Capture                | Ingress Tool Transfer            |                                              | Endpoint Denial of Service   |
| Replication Through Removable Media |                                         | Hijack Execution Flow (0/1)          |                                         | Indicator Removal on Host (0/2)       | Steal Application Access Token (0/1)  | System Information Discovery                 |                                     | Call Control                 | Non-Standard Port                |                                              | Generate Traffic from Victim |
| SIM Card Swap                       |                                         | Linked Devices                       |                                         | Input Injection                       | Masquerading (0/1)                    | System Network Configuration Discovery (0/2) |                                     | Clipboard Data               | Out of Band Data                 |                                              | Input Injection              |
| Supply Chain Compromise (0/3)       |                                         | Scheduled Task/Job                   |                                         | Masquerading (0/1)                    |                                       | System Network Connections Discovery         |                                     | Data from Local System       | Remote Access Software           |                                              | Network Denial of Service    |
|                                     |                                         |                                      |                                         | Native API                            |                                       |                                              |                                     | Input Capture (0/2)          | Web Service (0/3)                |                                              | SMS Control                  |
|                                     |                                         |                                      |                                         | Obfuscated Files or Information (0/2) |                                       |                                              |                                     | Linked Devices               |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Process Injection (0/2)               |                                       |                                              |                                     | Location Tracking (0/2)      |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Proxy Through Victim                  |                                       |                                              |                                     | Protected User Data (0/1)    |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Subvert Trust Controls (0/1)          |                                       |                                              |                                     | Screen Capture               |                                  |                                              |                              |
|                                     |                                         |                                      |                                         | Virtualization Solution               |                                       |                                              |                                     |                              |                                  |                                              |                              |

score 7

legend

## Step F: Annotations

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layer x +

Selection Controls Layer Controls Technique Controls

| Initial Access<br>8 techniques      | Execution<br>4 techniques               | Persistence<br>8 techniques          | Privilege Escalation<br>3 techniques    | Defense Evasion<br>17 techniques      | Credential Access<br>6 techniques     | Discovery<br>8 techniques                    | Lateral Movement<br>2 techniques    | Collection<br>15 techniques  | Command and Control<br>9 techniques | Malicious apps may                 |
|-------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------------|-------------------------------------|------------------------------|-------------------------------------|------------------------------------|
| Application Versioning              | Command and Scripting Interpreter (0/1) | Boot or Logon Initialization Scripts | Abuse Elevation Control Mechanism (0/1) | Application Versioning                | Abuse Accessibility Features          | File and Directory Discovery                 | Exploitation of Remote Services     | Abuse Accessibility Features | Application Layer Protocol (0/1)    | Exfiltration Over C2 Channel (0/1) |
| Drive-By Compromise                 | Exploitation for Client Execution       | Compromise Application Executable    | Exploitation for Privilege Escalation   | Download New Code at Runtime          | Access Notifications                  | Location Tracking (0/2)                      | Replication Through Removable Media | Access Notifications         | Call Control                        | Call Control                       |
| Exploitation for Initial Access     | Native API                              | Compromise Client Software Binary    | Process Injection (0/1)                 | Execution Guardrails (0/1)            | Clipboard Data                        | Network Service Scanning                     |                                     | Adversary-in-the-Middle      | Dynamic Resolution (0/1)            | Data Destruction                   |
| Lockscreen Bypass                   | Scheduled Task/Job                      | Event Triggered Execution (0/1)      |                                         | Foreground Persistence                | Credentials from Password Store (0/1) | Process Discovery                            |                                     | Archive Collected Data       | Encrypted Channel (0/3)             | Data Encrypted for Impact          |
| Phishing                            |                                         | Foreground Persistence               |                                         | Process Injection (T1631) (0/1)       | Input Capture (0/2)                   | Software Discovery (0/1)                     |                                     | Audio Capture                | Ingress Tool Transfer               | Data Manipulation (0/1)            |
| Replication Through Removable Media |                                         | Hijack Execution Flow (0/1)          |                                         | Indicator Removal on Host (0/3)       | Steal Application Access Token (0/1)  | System Information Discovery                 |                                     | Clipboard Data               | Non-Standard Port                   | Endpoint Denial of Service         |
| SIM Card Swap                       |                                         | Linked Devices                       |                                         | Input Injection                       |                                       | System Network Configuration Discovery (0/2) |                                     | Data from Local System       | Out of Band Data                    | Generate Traffic from Victim       |
| Supply Chain Compromise (0/3)       |                                         | Scheduled Task/Job                   |                                         | Masquerading (0/1)                    |                                       | System Network Connections Discovery         |                                     | Input Capture (0/2)          | Remote Access Software              | Input Injection                    |
|                                     |                                         |                                      |                                         | Native API                            |                                       |                                              |                                     | Linked Devices (0/3)         | Web Service (0/3)                   | Network Denial of Service          |
|                                     |                                         |                                      |                                         | Obfuscated Files or Information (0/2) |                                       |                                              |                                     | Location Tracking (0/2)      |                                     | SMS Control                        |
|                                     |                                         |                                      |                                         | Process Injection (0/1)               |                                       |                                              |                                     | Protected User Data (0/1)    |                                     |                                    |
|                                     |                                         |                                      |                                         | Proxy Through Victim                  |                                       |                                              |                                     | Screen Capture               |                                     |                                    |
|                                     |                                         |                                      |                                         | Subvert Trust Controls (0/1)          |                                       |                                              |                                     |                              |                                     |                                    |
|                                     |                                         |                                      |                                         | Virtualization Solution               |                                       |                                              |                                     |                              |                                     |                                    |

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## Step G: Export

ATT&CK v19 has been released! Check out the [blog post](#) for more information. | MITRE ATT&CK®

layer x +

Selection Controls Layer Controls Technique Controls

Export

render layer to SVG

| Initial Access<br>8 techniques      | Execution<br>4 techniques               | Persistence<br>8 techniques          | Privilege Escalation<br>3 techniques    | Defense Evasion<br>17 techniques      | Credential Access<br>6 techniques     | Discovery<br>8 techniques                    | Lateral Movement<br>2 techniques    | Collection<br>15 techniques  | Command and Control<br>9 techniques | Malicious apps may                 |
|-------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------------|-------------------------------------|------------------------------|-------------------------------------|------------------------------------|
| Application Versioning              | Command and Scripting Interpreter (0/1) | Boot or Logon Initialization Scripts | Abuse Elevation Control Mechanism (0/1) | Application Versioning                | Abuse Accessibility Features          | File and Directory Discovery                 | Exploitation of Remote Services     | Abuse Accessibility Features | Application Layer Protocol (0/1)    | Exfiltration Over C2 Channel (0/1) |
| Drive-By Compromise                 | Exploitation for Client Execution       | Compromise Application Executable    | Exploitation for Privilege Escalation   | Download New Code at Runtime          | Access Notifications                  | Location Tracking (0/2)                      | Replication Through Removable Media | Access Notifications         | Call Control                        | Call Control                       |
| Exploitation for Initial Access     | Native API                              | Compromise Client Software Binary    | Process Injection (0/1)                 | Execution Guardrails (0/1)            | Clipboard Data                        | Network Service Scanning                     |                                     | Adversary-in-the-Middle      | Dynamic Resolution (0/1)            | Data Destruction                   |
| Lockscreen Bypass                   | Scheduled Task/Job                      | Event Triggered Execution (0/1)      |                                         | Foreground Persistence                | Credentials from Password Store (0/1) | Process Discovery                            |                                     | Archive Collected Data       | Encrypted Channel (0/3)             | Data Encrypted for Impact          |
| Phishing                            |                                         | Foreground Persistence               |                                         | Hooking                               | Input Capture (0/2)                   | Software Discovery (0/1)                     |                                     | Audio Capture                | Ingress Tool Transfer               | Data Manipulation (0/1)            |
| Replication Through Removable Media |                                         | Hijack Execution Flow (0/1)          |                                         | Indicator Removal on Host (0/3)       | Steal Application Access Token (0/1)  | System Information Discovery                 |                                     | Clipboard Data               | Non-Standard Port                   | Endpoint Denial of Service         |
| SIM Card Swap                       |                                         | Linked Devices                       |                                         | Input Injection                       |                                       | System Network Configuration Discovery (0/2) |                                     | Data from Local System       | Out of Band Data                    | Generate Traffic from Victim       |
| Supply Chain Compromise (0/3)       |                                         | Scheduled Task/Job                   |                                         | Masquerading (0/1)                    |                                       | System Network Connections Discovery         |                                     | Input Capture (0/2)          | Remote Access Software              | Input Injection                    |
|                                     |                                         |                                      |                                         | Native API                            |                                       |                                              |                                     | Linked Devices (0/3)         | Web Service (0/3)                   | Network Denial of Service          |
|                                     |                                         |                                      |                                         | Obfuscated Files or Information (0/2) |                                       |                                              |                                     | Location Tracking (0/2)      |                                     | SMS Control                        |
|                                     |                                         |                                      |                                         | Process Injection (0/1)               |                                       |                                              |                                     | Protected User Data (0/1)    |                                     |                                    |
|                                     |                                         |                                      |                                         | Proxy Through Victim                  |                                       |                                              |                                     | Screen Capture               |                                     |                                    |
|                                     |                                         |                                      |                                         | Subvert Trust Controls (0/1)          |                                       |                                              |                                     |                              |                                     |                                    |
|                                     |                                         |                                      |                                         | Virtualization Solution               |                                       |                                              |                                     |                              |                                     |                                    |

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[illegible]

- End of Lab Task -